been satisfactorily explained. The old phases of ablaut have been preserved in the present and preterite of a few Gothic verbs, as **lētan,** *to let,* **laílōt, laílōtum, lētans; saian,** *to sow,* **saísō, saí·sō·um, saians.**

§ **228.** The ablaut-series as given in § **226** have for practical reasons been limited to the phases of ablaut as they appear in the various classes of strong verbs. From an Indo-Germanic point of view, the series I–V belong to one and the same series, generally called the **e**-series, which underwent in primitive Germanic various modifications upon clearly defined lines. What is called the sixth ablaut-series in the Germanic languages is really a mixture of several original series, owing to several Indg. vowel-sounds having fallen together in prim. Germanic; thus the **a,** which occurs in the present and past participle, corresponds to three Indg. vowels, viz. **a** (§ **17**), **o** (§ **20**), and **ə** (§ **22**); and the **ō** in the preterite corresponds to Indg. **ā** (§ **23**), and Indg. **ō** (§ **26**). For the phases of ablaut which do not occur in the various parts of strong verbs; and for traces of ablaut-series other than those given above, the student should consult Brugmann's *Kurze vergleichende Grammatik der indogermanischen Sprachen,* pp. 138–50.

## CHAPTER VIII

### THE FIRST SOUND-SHIFTING, VERNER'S LAW, AND OTHER CONSONANT CHANGES WHICH TOOK PLACE IN THE PRIMITIVE GERMANIC LANGUAGE

§ **229.** The first sound-shifting, popularly called Grimm's Law, refers to the changes which the Indo-Germanic explosives underwent in the period of the Germanic primitive community, i. e. before the Germanic parent language became differentiated into the separate Germanic lan-

guages :—Gothic, O. Norse, O. English, O. Frisian, O. Saxon (= O. Low German), O. Low Franconian (O. Dutch), and O. High German.

The Indo-Germanic parent language had the following system of consonants :—

| | | Labial. | Dental. | Palatal. | Velar. |
|---|---|---|---|---|---|
| *Explosives* | tenues | p | t | k | q |
| | mediae | b | d | g | g |
| | tenues aspiratae | ph | th | kh | qh |
| | mediae aspiratae | bh | dh | gh | gh |
| *Spirants* | voiceless | | s | | |
| | voiced | | z | j | |
| *Nasals* | | m | n | ń | ŋ |
| *Liquids* | | | l, r | | |
| *Semivowels* | | w (u̯) | | j (i̯) | |

Note.—1. Explosives are consonants which are formed with complete closure of the mouth passage, and may be pronounced with or without voice, i. e. with or without the vocal cords being set in action; in the former case they are said to be voiced (e. g. the mediae), and in the latter voiceless (e. g. the tenues). The aspirates are pronounced like the simple tenues and mediae followed by an **h**, like the Anglo-Irish pronunciation of **t** in **tell**.

The palatal explosives are formed by the front or middle of the tongue and the roof of the mouth (hard palate), like **g**, **k** (**c**) in English **get**, **good**, **kid**, **could**; whereas the velars are formed by the root of the tongue and the soft palate (velum). The latter do not occur in English, but are common in Hebrew, and are often heard in the Swiss pronunciation of German. In the parent Indo-Germanic language there were two kinds of velars, viz. pure velars and velars with lip rounding. The pure velars fell together with the Indg. palatals in Germanic, Latin, Greek, and Keltic, but were kept apart in the Aryan and Baltic-Slavonic languages. The velars with lip rounding appear in the Germanic languages partly with and partly without labialization, see § **237**. The palatal and velar nasals only

occurred before their corresponding explosives, **ńk, ńg**; **ŋq, ŋg**, &c.

2. Spirants are consonants formed by the mouth passage being narrowed at one spot in such a manner that the outgoing breath gives rise to a frictional sound at the narrowed part.

**z** only occurred before voiced explosives, e. g. ***nizdos** = Lat. **nīdus**, English **nest**; ***ozdos** = Gr. **ὄζος**, Germ. **ast**, *bough*.

**j** was like the widely spread North German pronunciation of **j** in **ja**, not exactly like the **y** in English **yes**, which is generally pronounced without distinct friction. **j** occurred very rarely in the prim. Indo-Germanic language. In the Germanic, as in most other Indo-Germanic languages, the frictional element in this sound became reduced, which caused it to pass into the so-called semivowel.

3. The nasals and liquids had the functions both of vowels and consonants (§ **16**).

4. The essential difference between the so-called semivowels and full vowels is that the latter always bear the stress (accent) of the syllable in which they occur, e. g. in English **ców, stáin** the first element of the diphthong is a vowel, the second a consonant; but in words like French **rwá** (written **roi**), **bjḗr** (written **bière**), the first element of the diphthong is a consonant, the second a vowel. In consequence of this twofold function, a diphthong may be defined as the combination of a sonantal with a consonantal vowel. And it is called a falling or rising diphthong according as the stress is upon the first or second element. In this book the second element of diphthongs is written **i, u** when the first element is the bearer of the stress, thus **ái, áu**, &c, but when the second element has the stress, the first element is written **j, w**, thus **já, wá**, &c.

5. In the writing down of prim. Germanic forms the signs **þ** (= **th** in Engl. **thin**), **đ** (= **th** in Engl. **then**), **ƀ** (= a bilabial spirant, which may be pronounced like the **v** in Engl. **vine**), **ʒ** (= **g** often heard in German **sagen**), **χ** (= NHG. **ch** and the **ch** in Scotch **loch**).

§ **230.** In the following tables of the normal equivalents of the Indg. explosives in Latin, Greek, and the Germanic languages, Table I contains the Indg. tenues **p, t, k,** the

mediae **b, d, g** and the pure velars **q, ǵ**. Table II contains the Indg. mediae aspiratae and the velars **q, ǵ** with labialization. The equivalents in the Germanic languages do not contain the changes caused by Verner's Law, &c. The East Franconian dialect is taken as the normal for OHG.

The following points should be noticed :—

(1) The Indg. tenues **p, t, k** and the mediae **b, d, g** generally remained unchanged in Latin and Greek.

(2) The pure velars (**q, ǵ**) fell together with the palatals **k, g** in Latin and Greek. They became **χ, k** in prim. Germanic, and thus fell together with the **χ, k** from Indg. **k, g**.

(3) The pure velar **ǵh** fell together with the original palatal **gh** in Latin and Greek.

(4) The Indg. mediae aspiratae became in prehistoric Latin and Greek tenues aspiratae, and thus fell together with the original tenues aspiratae.

(5) The Indg. tenues aspiratae became voiceless spirants in prim. Germanic, and thus fell together with the voiceless spirants from the Indg. tenues. See § **233**.

(6) In Latin Indg. **q** with labialization became **qu**, rarely **c**. **ǵ** with labialization became **v** (but **gu** after **n**, and **g** when the labialized element had been lost, as **gravis** = Gr. βαρύς, *heavy*).

Indg. **ph, bh** became **f** initially and **b** medially.

Indg. **th, dh** became **f** initially, **b** medially before and after **r**, before **l** and after **u** (**w**), in other cases **d**.

Indg. **kh, gh** became **h** initially before and medially between vowels; **g** before and after consonants, and **f** before **u** (**w**).

Indg. **qh, ǵh** with labialization became **f** initially, **v** medially except that after **n** it became **gu**.

(7) In Greek Indg. **q, ǵ** with labialization became **π, β**

before non-palatal vowels (except **u**) and before consonants (except Indg. **j**); τ, δ before palatal vowels; and κ, γ before and after **u**.

Indg. **ph, bh** became φ; **th, dh** became θ; and **kh, gh** became χ.

Indg. **qh, gh** with labialization became φ before non-palatal vowels (except **u**) and before consonants (except Indg. **j**); θ before palatal vowels; and χ before and after **u**.

(8) When two consecutive syllables would begin with aspirates, the first was de-aspirated in prehistoric times in Sanskrit and Greek, as Skr. **bándhanam,** *a binding,* Goth. OE. **bindan,** OHG. **bintan,** *to bind*; Skr. **bṓdhati,** *he learns, is awake,* Gr. πεύθεται, *he asks, inquires,* Goth. **anabiudan,** OE. **bēodan,** *to bid,* OHG. **biotan,** *to offer,* root **bheudh-**; Gr. κανθύλη, *a swelling,* OE. **gund,** OHG. **gunt,** *matter, pus*; Gr. θρίξ, *hair,* gen. τρίχος; ἔχω, *I have,* fut. ἕξω.

(9) In OHG. the prim. Germanic explosives **p, t** became the affricatae **pf, tz** (generally written **zz, z**), initially, as also medially after consonants, and when doubled. But prim. Germanic **p, t, k** became the double spirants **ff, ȝȝ, hh** (also written **ch**) medially between vowels and finally after vowels. The double spirants were simplified to **f, ȝ, h** when they became final or came to stand before other consonants, and also generally medially when preceded by a long vowel or diphthong.

TABLE I.

| Indg. | Latin | Greek | P. Germanic | Gothic | OE. | OHG. |
|---|---|---|---|---|---|---|
| p | p | π | f | f | f | f |
| t | t | τ | þ | þ | þ | d |
| k, q | c | κ | χ | h, χ | h, χ | h, χ |
| b | b | β | p | p | p | pf, ff |
| d | d | δ | t | t | t | z, zz |
| g, ǵ | g | γ | k | k | c | k, hh |

TABLE II.

| Indg. | Latin | Greek | P. Germanic | Gothic | OE. | OHG. |
|---|---|---|---|---|---|---|
| q | qu, c | π, τ, κ | χw, χ | hw, h | hw, h | (h)w, h |
| ǵ | v, gu, g | β, δ, γ | kw, k | q, k | cw, c | qu; k, hh |
| bh | f, b | φ | b, ƀ | b, ƀ | b, ƀ, (f) | b |
| dh | f, b, d | θ | d, đ | d, đ | d | t |
| gh | h, g, f | χ | g, ʒ | g, ʒ | g, ʒ | g |
| ǵh | f, v, gu | φ, θ, χ | ʒw, ʒ, w | ʒ, w | g, ʒ, w | g, w |

§ **231.** The Indg. tenues **p, t, k, q** became in prim. Germanic the voiceless spirants **f, þ, χ, χ (χw)**.

**p > f.** Lat. **pēs,** Gr. πούς, OE. OS. **fōt,** Goth. **fōtus,** O.Icel. **fōtr,** OHG. **fuoz,** *foot*; Lat. **piscis,** OE. **fisc,** Goth. **fisks,** O.Icel. **fiskr,** OS. OHG. **fisk,** *fish*; Gr. πλωτός, *floating, swimming,* OE. OS. **flōd,** Goth. **flōdus,** O.Icel. **flōð,** OHG **fluot,** *flood, tide*; Lat. **pecu,** OE. **feoh,** Goth. **faíhu,** O.Icel. **fē,** OS. **fehu,** OHG. **fihu,** *cattle*; Lat. **nepos,** OE. **nefa,** O.Icel. **nefe,** OHG. **nefo,** *nephew.*

**t > þ.** Lat. **tu,** Gr. Doric τύ, OE. O.Icel. OS. **þū,** Goth. **þu,** OHG. **dū,** *thou*; Lat. **trēs,** Gr. τρεῖς, OE. OS. **þrī,** O.Icel. **þrīr,** OHG. **drī,** *three*; Lat. **tenuis,** OE. **þynne,** O.Icel. **þunnr,** OHG. **dunni,** *thin*; O.Lat. **tongēre,** *to know,* OE. **þencan,** Goth. **þagkjan,** OS. **þenkian,** OHG. **denchen,** *to think*; Lat. **frāter,** OE. **brōþor,** Goth. **brōþar,** O.Icel. **brōðer,** OS. **brōđar,** OHG. **bruoder,** *brother*; Lat. **vertō,** *I turn,* OE. **weorþan,** Goth. **waírþan,** O.Icel. **verða,** OS. **werđan,** OHG. **werdan,** to become.

**k > χ.** Lat. **canis,** Gr. κύων, OE. OS. **hund,** Goth. **hunds,** O.Icel. **hundr,** OHG. **hunt,** *hound, dog*; Lat. **cor** (gen. **cordis**), Gr. καρδίᾱ, OE. **heorte,** Goth. **haírtō,** O.Icel. **hjarta,** OS. **herta,** OHG. **herza,** *heart*; Lat. **centum,** Gr. ἑ-κατόν, OE. Goth. OS. **hund,** OHG. **hunt,** *hundred*; Lat. **pecu,** OE. **feoh,** Goth. **faíhu,** O.Icel. **fē,** OS. **fehu,** OHG. **fihu,** *cattle*; Lat. **decem,** Gr. δέκα, OE. **tīen** from ***teohuni-,** older ***teχuni-** (cp. §§ **87, 447**), Goth. **taíhun,** OS. **tehan,** OHG. **zehan,** *ten*; Lat. **dūcō,** *I lead,* OE. **tēon** from ***tēohan,** older ***teuχan** (§ **137**), Goth. **tiuhan,** OS. **tiohan,** OHG. **ziohan,** *to draw, lead*

**q > χ (χw).** Lat. **capiō,** *I take, seize,* OE. **hebban,** Goth. **hafjan,** O.Icel. **hefja,** OS. **hebbian,** OHG. **heffen,** *to raise*; Lat. **canō,** *I sing,* OE. **hana, hona,** Goth. **hana,** O Icel. **hane,** OS. OHG. **hano,** *cock,* lit. *singer*; Lat. **vincō** (perf. **vīcī**), *I conquer,* Goth. **weihan,** OHG. **wīhan,** *to fight.*

Lat. **quis,** Gr. τίς, OE. **hwā,** Goth. **hwas,** OS. **hwē,** OHG.

**hwer,** *who?*; Lat. **linquō,** Gr. λείπω, *I leave,* OE. **līon, lēon** from ***līohan,** older ***līχwan** (see §§ **127, 246**), OS. OHG. **līhan,** *to lend.*

NOTE.—1. The Indg tenues remained unshifted in the combination **s** + tenuis.

**sp.** Lat. **spuere,** OE. OS. OHG. **spīwan,** Goth. **speiwan,** *to vomit, spit*; Lat. **con-spiciō,** *I look at,* OHG. **spehōn,** *to spy.*

**st.** Gr. στείχω, *I go,* Lat. **vestīgium,** *footstep,* OE. OS. OHG. **stīgan,** Goth. **steigan,** O.Icel. **stīga,** *to ascend*; Lat. **hostis,** *stranger, enemy,* OE. **giest,** Goth. **gasts,** O.Icel. **gestr,** OS. OHG. **gast,** *guest.*

**sk.** Gr. σκιά, *shadow,* OE. OS. OHG. **scīnan,** Goth. **skeinan,** O.Icel. **skīna,** *to shine*; Lat. **piscis,** OE. **fisc,** Goth. **fisks,** O.Icel. **fiskr,** OS. OHG. **fisk,** *fish.*

**sq.** Gr. θυο-σκόος, *sacrificing priest,* OE. **scēawian,** Goth. ***skaggwōn,** OS. **scauwōn,** OHG. **scouwōn,** *to look, view.*

2. The **t** also remained unshifted in the Indg. combinations **pt, kt, qt.**

**pt > ft.** Lat. **neptis,** OE. OHG. **nift,** *niece, granddaughter*; Lat. **captus,** *a taking, seizing,* OE. **hæft,** OHG. **haft,** *one seized or taken, a captive.*

**kt > χt.** Lat. **octō,** Gr. ὀκτώ, OE. **eahta,** Goth. **ahtáu,** OS. OHG. **ahto,** *eight*; Gr. ὀ-ρεκτός, *stretched out,* Lat. **rēctus,** OE. **riht,** Goth. **raíhts,** OS. OHG. **reht,** *right, straight.*

**qt > χt.** Gen. sing. Lat **noctis,** Gr. νυκτός, nom. OE. **neaht, niht,** Goth. **nahts,** OS. OHG. **naht,** *night.*

§ **232.** The Indg. mediae **b, d, g, ǵ** became the tenues **p, t, k, k (kw).**

**b > p.** Lithuanian **dubùs,** OE. **dēop,** Goth. **diups,** O.Icel. **djūpr,** OS. **diop,** OHG. **tiof,** *deep*; Lithuanian **trobà,** *house,* OE. **þorp,** OS. **thorp,** OHG. **dorf,** *village,* Goth. **þaúrp,** *field*; O. Bulgarian **slabŭ,** *slack, weak,* OE. **slǣpan,** Goth. **slēpan,** OS. **slāpan,** OHG. **slāfan,** *to sleep,* originally, *to be slack.* **b** was a rare sound in the parent language.

**d > t.** Lat. **decem,** Gr. δέκα, OE **tīen,** Goth. **taíhun,** O.Icel. **tīo,** OS. **tehan,** OHG. **zehan,** *ten*; Lat. **dens** (gen. **dentis**), OE. **tōþ,** Goth. **tunþus,** OS. **tand,** OHG. **zand,** *tooth*;

Lat. **vidēre,** *to see,* OE. Goth. OS. **witan,** O.Icel. **vita,** OHG. **wiȝȝan,** *to know*; Lat. **edō,** Gr. ἔδω, *I eat,* OE. OS. **etan,** Goth. **itan,** O.Icel. **eta,** OHG. **eȝȝan,** *to eat*; gen. Lat. **pedis,** Gr. ποδός, nom. OE. OS. **fōt,** Goth. **fōtus,** O.Icel. **fōtr,** OHG. **fuoȝ,** *foot.*

**g > k.** Lat. **genu,** Gr. γόνυ, OE. **cnēo,** Goth. **kniu,** O.Icel. **knē,** OS. OHG. **knio,** *knee*; Lat. **gustō,** *I taste,* Gr. γεύω, *I let taste,* OE. **cēosan,** Goth. **kiusan,** O.Icel. **kjōsa,** OS. OHG. **kiosan,** *to test, choose*; Lat. **ager,** Gr. ἀγρός, OE. **æcer,** Goth. **akrs,** O.Icel. **akr,** *field, acre*; Lat. **egō,** Gr. ἐγώ, OE. **ic,** Goth. OS. **ik,** O.Icel. **ek,** OHG. **ih,** *I.*

**ǥ > k (kw).** Lat. **gelu,** *frost,* OE. **ceald,** Goth. **kalds,** O.Icel. **kaldr,** OS. **kald,** OHG. **kalt,** *cold*; Lat. **augēre,** Goth. **áukan,** O.Icel. **auka,** OS. **ōkian,** OHG. **ouhhōn,** *to add, increase,* OE. participial adj. **ēacen,** *great*; Lat. **jugum,** Gr. ζυγόν, OE. **geoc,** Goth. **juk,** OHG. **joh,** *yoke.*

Gr. Bœotian βανά, OE. **cwene,** Goth. **qinō,** OS. **quena,** *woman, wife*; Gr. βίος from ***ǥĭwos,** *life,* Lat. **vīvos** from ***gwīwos,** OE. **cwicu,** Goth. **qius,** O.Icel. **kvikr,** OS. **quik,** OHG. **quec,** *quick, alive*; Lat. **veniō** from ***gwemjō,** *I come,* Gr. βαίνω from *βαjνω, older *βαμjω = Indg. ***ǥmjō,** *I go,* OE. OS. **cuman,** Goth. **qiman,** O.Icel. **koma,** OHG. **queman,** *to come.*

§ **233.** The Indg. tenues aspiratae became voiceless spirants in prim. Germanic, and thus fell together with and underwent all further changes in common with the voiceless spirants which arose from the Indg. tenues (§ **231**), the latter having also passed through the intermediate stage of tenues aspiratae before they became spirants. The tenues aspiratae were, however, of so rare occurrence in the prim. Indg. language that two or three examples must suffice for the purposes of this book; for further examples and details, the learner should consult Brugmann's *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen,* vol. I :—Skr. root **sphal-,** *run*

*violently against*, OE. **feallan**, O.Icel. **falla**, OS. OHG. **fallan**, *to fall*; Gr. ἀ-σκηθής, *unhurt*, OE. **sceþþan**, Goth. **skaþjan**, OHG. **skadōn**, *to injure*; Gr. σχίζω, *I split*, OE. **scādan**, Goth. **skáidan**, OHG. **sceidan**, *to divide, separate*; Gr. φάλλη, OE. **hwæl**, O.Icel. **hvalr**, OHG. **(h)wal**, *whale*.

§ **234.** The Indg. mediae aspiratae probably became first of all the voiced spirants **ƀ, đ, ʒ, ʒ(w)**. These sounds underwent the following changes during the prim. Germanic period:—**ƀ, đ** initially, and **ƀ, đ, ʒ** medially after their corresponding nasals, became the voiced explosives **b, d, g**, as

**b.** OE. OS. OHG. **beran**, Goth. **baíran**, O.Icel. **bera**, *to bear*, Skr. **bhárāmi**, Gr. φέρω, Lat. **ferō**, *I bear*; OE. OS. **bītan**, Goth. **beitan**, O.Icel. **bīta**, OHG. **bīʒʒan**, *to bite*, Skr. **bhédāmi**, Lat. **findō**, *I cleave*; OE. **brōþor**, Goth. **brōþar**, O.Icel. **brōðer**, OS. **brōđar**, OHG. **bruoder**, Skr. **bhrā́tar-**, Lat. **frāter**, *brother*, Gr. φρᾱ́τηρ, *member of a clan*.

OE. **ymbe**, OS. OHG. **umbi**, Gr. ἀμφί, *around*; OE. **camb, comb**, O.Icel. **kambr**, OHG. **kamb**, *comb*, Skr. **jámbhas**, *tooth*, Gr. γόμφος, *bolt, nail*, prim. form ***gombhos**.

**d.** OE. **dæg**, Goth. **dags**, O.Icel. **dagr**, OS. **dag**, OHG. **tag**, *day*, Skr. **ni-dāghás**, older ***ni-dhāghás**, *hot season, summer*, Indg. form ***dhoghos**; OE. **dǣd**, OS. **dād**, OHG. **tāt**, *deed*, related to Gr. θή-σω, *I shall place*, Skr. **dhā́ma**, *law, dwelling-place*, root **dhē-**, *put, place*; OE. **dohtor**, Goth. **daúhtar**, OS. **dohter**, OHG. **tohter**, Gr. θυγάτηρ, *daughter*; OE. **duru**, OS. **duri**, OHG. **turi**, Gr. θύρᾱ, *door*.

OE. Goth. OS. **bindan**, O.Icel. **binda**, OHG. **bintan**, *to bind*, Skr. **bándhanam**, *a binding*, cp. πενθερός, *father-in-law*, Lat. **of-fendimentum**, *chin-cloth*, root **bhendh-**.

**g.** OE. **enge**, Goth. **aggwus**, OS. OHG. **engi**, *narrow*, cp. Lat. **angō**, Gr. ἄγχω, *I press tight*, root **aṅgh-**; OE. **lang, long**, Goth. **laggs**, O.Icel. **langr**, OS. OHG. **lang**, Lat. **longus**, *long*.

§ **235.** **ƀ, đ, ʒ** remained in other positions, and their further development belongs to the history of the separate

Germanic languages. In Goth. **ƀ, đ** (written **b, d**) remained medially after vowels, but became explosives (**b, d**) after consonants. They became **f, þ** finally after vowels and before final **-s**. **ʒ** remained medially between vowels, and medially after vowels before voiced consonants, but became **χ** (written **g**) finally after vowels and before final **-s**. It became **g** initially, and also medially after voiced consonants. In O.Icel. **ƀ** (written **f**) remained medially between and finally after voiced sounds, but became **f** before voiceless sounds. **đ** (written **ð**) generally remained medially and finally. **ʒ** remained medially after vowels and liquids, but became **χ** and then disappeared finally. It became **g** initially. **đ** became **d** in all the West Germanic languages and then **d** became **t** in OHG. In OS. **ƀ** (written **ƀ, b**) generally remained between voiced sounds. It became **f** medially before **l** and **n**, and before voiceless consonants, and also finally. **ʒ** (written **g**) remained initially and medially, but became **χ** finally, although it was generally written **g**. In OHG. **ƀ, ʒ** became **b, g**. On the history of **ƀ, ʒ** in OE. see §§ **293–4, 313–24**. Geminated **ƀƀ, đđ, ʒʒ**, of whatever origin, became **bb, dd, gg** in the prehistoric period of all the Germanic languages. Examples are :—Goth. ***nibls**, OS. **neƀal**, OHG. **nebul**, Lat. **nebula**, Gr. **νεφέλη**, *mist, cloud*, cp. Skr. **nábhas**, Gr. **νέφος**, *cloud*; OE. **lēof**, Goth. **liufs**, O.Icel. **ljūfr**, OS. **liof**, OHG. **liob**, *dear*, original form ***leubhos**, cp. Skr. **lúbhyāmi**, *I feel a strong desire*, Lat. **lubet** (**libet**), *it pleases*; OE. OS. **ūder**, OHG. **ūter**, Skr. **ūdhar**, Gr. **οὖθαρ**, *udder*; OE. **rēad**, Goth **ráuþs**, O.Icel. **rauðr**, OS. **rōd**, OHG. **rōt**, prim. form ***roudhos**, cp. Skr. **rudhirás**, Gr. **ἐ-ρυθρός**, prim. form ***rudhros**, *red*; OE. Goth. **guma**, O.Icel. **gume**, OS. OHG. **gumo**, Lat. **homō**, prim. stem-form ***ghomon-**, *man*; OE. **gōs**, O.Icel. **gās**, OHG. **gans**, Gr. **χήν**, *goose*; OE. OS. OHG. **wegan**, Goth. **ga-wigan**, O.Icel. **vega**, *to move, carry*, Lat. **vehō**, prim. form ***weghō**, *I carry*; OE.

**giest,** Goth. **gasts,** O.Icel. **gestr,** OS. OHG. **gast,** *guest,* Lat. **hostis,** *stranger, enemy,* prim. form ***ǥhostis**; OE. OS. OHG. **stīgan,** Goth. **steigan,** O.Icel. **stīga,** *to ascend,* Gr. στείχω, prim. form ***steiǥhō,** *I go,* cp. Lat. **vestīgium,** *footstep.*

§ **236.** Various theories have been propounded as to the chronological order in which the Indg. tenues, tenues aspiratae, mediae, and mediae aspiratae, were changed by the first sound-shifting in prim. Germanic. But not one of these theories is satisfactory. Only so much is certain that at the time when the Indg. mediae became tenues, the Indg. tenues must have been on the way to becoming voiceless spirants, otherwise the two sets of sounds would have fallen together.

§ **237.** We have already seen (§ **230**) that the parent Indg. language contained two series of velars: (1) pure velars which never had labialization. These velars fell together with the palatals in the Germanic, Latin, Greek, and Keltic languages, but were kept apart in the Aryan and Baltic-Slavonic languages. (2) Velars with labialization. These velars appear in the Germanic languages partly with and partly without labialization; in the latter case they fell together with prim. Germanic **χ, k, ȝ** which arose from Indg. **k, g, gh.** The most commonly accepted theory is that the Indg. labialized velars **q, ǥ, ǥh** regularly became **χ, k, ȝ** in prim. Germanic before Indg. **ŭ, ō, o** (= Germanic **a,** § **20**), and **χw, kw, ȝw** before Indg. **ĕ, ĭ, ə, a, ā** (= Germanic **ō,** § **23**); and that then the law became greatly obscured during the prim. Germanic period through form-transference and levelling out in various directions, as Goth. **qam,** OHG. **quam,** prim. form ***ǥoma,** *I came,* for Goth. OHG. ***kam** after the analogy of Goth. **qima,** OHG. **quimu,** original form ***ǥemō,** *I come*; Goth. **hvas,** *who?,* Indg. ***qos** for ***has** after the analogy of the gen. **hvis** = Indg. ***qeso,** &c.

NOTE.—In several words the Indg. velars, when preceded or followed by a **w** or another labial in the same word, appear in the Germanic languages as labials by assimilation. The most important examples are:—OE. OS. **wulf**, Goth. **wulfs**, OHG. **wolf** = Gr. λύκος, for *ϝλύκος, prim. form *wlqos, cp. Skr. **vṛ́kas**, *wolf*; OE. **fēower** (but **fyþer-fēte**, *four-footed*), Goth. **fidwōr**, OS. OHG. **fior**, prim. form *qetwṓres, cp. Lithuanian **keturì**, Lat. **quattuor**, Gr. τέσσαρες, Skr. **catvā́ras**, *four*; OE. OS. **fīf**, Goth. **fimf**, OHG. **fimf**, **finf**, prim. form *peŋqe, cp. Skr. **páṅca**, Gr. πέντε, Lat. **quīnque** (for *pīnque), *five*; OE. **weorþan**, Goth. **waírpan**, O.Icel. **verpa**, OS. **werpan**, OHG. **werfan**, *to throw*, cp. O. Bulgarian **vrĭgą**, *I throw*; OE. **swāpan**, OHG. **sweifan**, *to swing*, cp. Lithuanian **swaikstù**, *I become dizzy*.

## VERNER'S LAW.

§ **238.** After the completion of the first sound-shifting, and while the principal accent was not yet confined to the root-syllable, a uniform interchange took place between the voiceless and voiced spirants, which may be thus stated:—

The medial or final spirants **f, þ, χ, χw, s** regularly became **b̧, đ, g, gw, z** when the vowel next preceding them did not, according to the original Indg. system of accentuation, bear the principal accent of the word.

The **b̧, đ, g, gw** which thus arose from Indg. **p, t, k, q** underwent in the Germanic languages all further changes in common with the **b̧, đ, g, gw** from Indg. **bh, dh, gh, gh.**

Verner's law manifests itself most clearly in the various parts of strong verbs, where the infinitive, present participle, present tense, and preterite (properly perfect) singular had the principal accent on the root-syllable, but the indicative pret. plural, the pret. subjunctive (properly optative), and past participle had the principal accent on the ending, as prim. Germanic ***wérþō** > OE. **weorþe**, *I become* = Skr. **vártā-mi**, *I turn*; pret. indic. 3. sing. ***wárþi** > OE. **wearþ**, *he became* = Skr. **va-várta**, *has turned*; pret. 1. pers. pl. ***wurđumí** > OE. ***wurdum** (**wurdon** is the 3. pers. pl. used for all persons) = Skr. **va-vrtimá**, *we have*

*turned*; past participle ***wurđaná-** > OE. **worden** = Skr. **va-vrtāná-**; OS. **birid**, OHG. **birit** = Skr. **bhárati**, *he bears*; Goth. 2. sing. indic. passive **baíraza** = Skr. **bhárasē**; Goth. **baírand**, OHG. **berant** = Skr. **bháranti**, *they bear*; present participle OE. **berende**, Goth. **baírands**, O.Icel. **berande**, OS. **berandi**, OHG. **beranti**, Gr. gen. **φέροντος**. Or to take examples from noun-forms, &c., we have e. g. Skr. **pitár-**, Gr. **πατέρ-** = prim. Germanic ***fađér-**, OE. **fæder**, Goth. **fadar**, O.Icel. **faðer**, OS. **fader**, OHG. **fater**, *father*; Gr. **πλωτός**, *floating, swimming*, OE. OS. **flōd**, Goth. **flōdus**, O.Icel. **flōð**, OHG. **fluot**, *flood, tide*; Skr. **çatám**, Gr. **ἑ-κατόν**, Lat. **centum** = prim. Germanic ***χunđóm**, older ***χumđóm**, OE. Goth. OS. **hund**, OHG. **hunt**, *hundred*; Indg. ***swékuros**, Goth. **swaíhra**, OHG. **swehur**, OE. **swēor** (§ **329**), *father-in-law*, beside Gr. **ἑκυρᾶ**, OE. **sweger**, OHG. **swigar**, *mother-in-law*; Gr. **δέκα**, Goth. **taíhun**, OS. **tehan**, OHG. **zehan**, *ten*, beside Gr. **δεκάς**, OE. OS. **-tig**, OHG. **-zug**, Goth. pl. **tigjus**, *decade*; Skr. **saptá**, Gr. **ἑπτά**, OE. **seofon**, Goth. **sibun**, OS. **siþun**, OHG. **sibun**, *seven*; Gr. **νυός** from ***σνυσός**, OE. **snoru**, OHG. **snura**, *daughter-in-law*; OHG. **haso** beside OE. **hara**, *hare*; Goth. **áusō** beside OE. **ēare**, *ear*.

The combinations **sp, st, sk, ss, ft, fs, hs,** and **ht** were not subject to this law.

NOTE.—The prim. Germanic system of accentuation was like that of Sanskrit, Greek, &c., i. e. the principal accent could fall on any syllable; it was not until a later period of the prim. Germanic language that the principal accent was confined to the root-syllable. See § **9**.

§ **239**. From what has been said above it follows that the interchanging pairs of consonants due to Verner's law were in prim. Germanic: **f—ƀ, þ—đ, s—z, χ—ʒ, χw—ʒw**. They underwent various changes partly in prim. Germanic, partly in West Germanic, and partly in Old English. Already in prim. Germanic **ʒw** became **ʒ** before **u**, but

**w** in other cases (§ **241**); and **ŋʒ** became **ŋg**. In West Germanic **đ** became **d** (§ **253**); **z** became **r** medially and was dropped finally (§ **252**); **χw** became **χ** (§ **246**). In OE. the two sounds **f**—**ƀ** fell together in **ƀ** (written **f**) medially, and in **f** finally, see §§ **293–4, 296**; **χ** disappeared between vowels (§ **329**), when preserved it was written **h**; and **þ, s** became voiced between vowels, although the **þ, s** were preserved in writing. So that for OE. we have the following interchanging pairs of consonants :—

**þ—d** **s—r**
**h** or loss of **h** (= prim. Germ. **χ**)—**g** (§ **320**)
**h** or loss of **h** (= prim. Germ. **χw**)—**g, w** (= prim. Germ. **ʒw**)
**h** or loss of **h** (= prim. Germ. **ŋχ**, § **245**)—**ng**.

**þ—d.** **cweþan,** *to say,* **līþan,** *to go,* **snīþan,** *to cut;* pret. sing. **cwæþ, lāþ, snāþ**; pret. pl. **cwǣdon, lidon, snidon**; pp. **cweden, liden, sniden**; **cwide,** *saying, proverb*; **snide,** *incision*; **dēaþ,** *death,* beside **dēad,** *dead.*

**s—r.** **cēosan,** *to choose,* **drēosan,** *to fall,* **forlēosan,** *to lose;* pret. sing. **cēas, drēas, forlēas**; pret. pl. **curon, druron, forluron**; pp. **coren, droren, forloren**; **cyre,** *choice*; **dryre,** *fall*; **lyre,** *loss.*

**h—g.** **flēon** (OHG. **fliohan**), *to flee,* **slēan** (Goth. **slahan**), *to strike, slay,* **tēon** (Goth. **tiuhan**), *to draw, lead*; pret. sing. **flēah, tēah**; pret. pl. **flugon, slōgon, tugon**; pp. **flogen, slægen, togen**; **slaga,** *homicide*; **slege,** *stroke, blow*; **here-toga,** *leader of an army, duke.*

**h—g, w.** **sēon** (Goth. **saíƕan**), *to see;* pret. sing. **seah**; pret. pl. WS. **sāwon,** Anglian **sēgon**; pp. WS. **sewen,** Anglian **segen**; **sīon, sēon** (Goth. ***seiƕan,** OHG. **sīhan**), *to strain*; pret. sing. **sāh**; pp. **siwen, sigen**; **horh,** *dirt,* gen. **horwes.** See Note 1 below.

**h—ng.** **fōn** (Goth. **fāhan,** prim. Germ. ***faŋχanan,** § **245**), *to seize,* **hōn** (Goth. **hāhan,** prim. Germ. ***χaŋχanan**),

*to hang*; pret. pl. **fēngon, hēngon**; pp **fangen, hangen**; **feng,** *grasp, booty*; **hangian,** *to hang*; **þīon, þēon** (Goth. **þeihan,** prim. Germ. **þiŋχanan**), *to thrive*; pret. pl. **þungon**; pp. **þungen**; the usual pret. pl. **þigon,** pp. **þigen,** were new formations, see § **492.**

NOTE.—1. The results of the operation of Verner's law were often disturbed in OE. through the influence of analogy and levelling, e. g. the **þ, s** of the present and pret. singular were extended to the pret. plural and pp. in **ābrēoþan,** *to fail*; **mīþan,** *to avoid*; **wrīþan,** *to twist*; **ārīsan,** *to arise*; **genesan,** *to recover*; **lesan,** *to collect*; pret. pl. **ābruþon, miþon, wriþon, ārison, genǣson, lǣson**; pp. **ābroþen, miþen, wriþen, ārisen, genesen, lesen.** The **g** of the pret. plural was levelled out into the singular in **flōg,** *he flayed*; **hlōg,** *he laughed*; **lōg,** *he blamed*; **slōg,** *he slew*; **þwōg,** *he washed*, see § **509.** The **nd** of the pret. plural and pp. was extended to the present and pret. singular in **findan,** pret. sing. **fand.** The regular forms of this verb would be ***fīþan** (= Goth. **finþan,** OS. **fīþan**), *to find*; pret. sing. ***fōþ** (see § **61**), pret. pl. **fundon,** pp. **funden.** The WS. pret. pl. **sāwon,** *they saw*, had its **w** from the pp. **sewen,** and conversely the Anglian pp. **segen,** *seen*, had its **g** from the pret. plural **sēgon,** see § **241.**

2. Causative verbs had originally suffix accentuation, and therefore also exhibit the change of consonants given above, as **weorþan,** *to become*: **ā-wierdan,** *to destroy, injure*, cp. Skr. **vártāmi,** *I turn*: **vartáyāmi,** *I cause to turn*; **līþan,** *to go*: **lǣdan,** *to lead*; **ā-rīsan,** *to arise*: **rǣran,** *to raise*; **genesan,** *to recover*: **nerian,** *to save*.

## OTHER CONSONANT CHANGES.

§ **240.** Most of the sound changes comprised under this paragraph might have been disposed of in the paragraphs dealing with the shifting of the Indg. mediae and mediae aspiratae, but to prevent any possible misunderstanding or confusion, it was thought advisable to reserve them for a special paragraph.

The Indg. mediae and mediae aspiratae became tenues

before a suffixal **t** or **s** already in the pre-Germanic period:—

| | | | |
|---|---|---|---|
| bt, bht | pt | bs, bhs | ps |
| dt, dht | tt | ds, dhs | ts |
| gt, ght | kt | gs, ghs | ks |
| ǵt, ǵht | qt | ǵs, ǵhs | qs |

Examples are: Lat. **nūptum, nūpsī,** beside **nūbere,** *to marry*; Skr. loc. pl. **patsú,** beside loc. sing. **padí,** *on foot*; Lat. **rēxī, rectum,** beside **regere,** *to rule*; Lat. **vēxī, vectum,** beside **vehere,** *to carry,* root **wegh-**; Lat. **lectus,** Gr. **λέχος,** *bed,* OE. **licgan,** Goth. **ligan,** *to lie down*; Skr. **yuktá-,** Gr. **ζευκτός,** Lat. **jūnctus,** *yoked,* root **jeug-,** cp. Skr. **yugám,** Gr. **ζυγόν,** Lat. **jugum,** OE. **geoc,** Goth. **juk,** *yoke*; &c.

Then **pt, kt, qt; ps, ks, qs** were shifted to **ft, χt; fs, χs** at the same time as the original Indg. tenues became voiceless spirants (§ **231**). And **tt, ts** became **ss** through the intermediate stage of **þt, þs** respectively. **ss** then became simplified to **s** after long syllables and before **r,** and then between the **s** and **r** there was developed a **t.**

This explains the frequent interchange between **p, ƀ** (b), and **f**; between **k, ȝ** (g), and **h** (i. e. χ); and between **t, þ, đ** (d), and **ss, s** in forms which are etymologically related.

**p, ƀ** (b)—**f.** OE. **scieppan,** Goth. **skapjan,** *to create,* beside OE. **ge-sceaft,** OHG. **gi-skaft,** *creature,* Goth. **ga-skafts,** *creation*; Goth. **giban,** OHG. **geban,** *to give,* beside Goth. **fra-gifts,** *a giving, espousal,* OE. OHG. **gift,** *gift*; OHG. **weban,** *to weave,* beside English **weft.**

**k, ȝ** (g)—**h.** OE. **wyrcan,** Goth. **waúrkjan,** OHG. **wurken,** *to work,* beside pret. and pp. OE. **worhte, worht,**

Goth. **waúrhta, waúrhts,** OHG. **worhta, giworht**; OE. **þyncan,** Goth. **þugkjan,** OHG. **dunken,** beside pret. and pp. OE. **þūhte, þūht,** Goth. **þūhta,** ***þūhts,** OHG. **dūhta, gidūht**; OE. **magon,** Goth. ***magun,** OHG. **magun,** *they may, can,* beside pret. OE. **meahta,** Goth. OHG. **mahta,** pp. Goth. **mahts,** cp. also OE. **meaht,** Goth. **mahts,** OHG. **maht,** *might, power*; OE. **bycgan,** Goth. **bugjan,** *to buy,* beside pret. and pp. OE. **bohte, boht,** Goth. **baúhta, baúhts**; OE. OHG. **bringan,** Goth. **briggan,** *to bring,* pret. and pp. OE. **brōhte, brōht,** OHG. **brāhta, gibrāht,** Goth. **brāhta,** ***brāhts.**

**t, þ, đ** (d)—**ss, s.** OE. Goth. OS. **witan,** O.Icel. **vita,** beside pret. OE. **wisse,** Goth. OS. OHG. **wissa,** O.Icel. **vissa,** participial adj. OE. **gewiss,** O.Icel. **viss,** OS. **wis(s),** OHG. **giwis(s),** *sure, certain*; OE. **sittan,** O.Icel. **sitja,** OS. **sittian,** *to sit,* beside OE. O.Icel. OS. **sess,** *seat*; OE. **cweþan,** Goth. **qiþan,** *to say,* beside Goth. **ga-qiss,** *consent*; Goth. **ana-biudan,** *to command,* beside **ana-busns,** *commandment,* pre-Germanic -***bhŭtsni-,** root **bheudh-.**

**ss** became **s** after long syllables and before **r**: OE. **hātan,** Goth. **háitan,** *to call,* beside OE. **hǣs** from ***haissi-,** *command*; OE. Goth. OS. **witan,** *to know,* beside OE. OS. OHG. **wīs,** *wise,* Goth. **unweis,** *unknowing*; OE. **etan,** Goth. **itan,** *to eat,* beside OE. **ǣs,** OHG. **ās,** *carrion.* OE. **fōstor,** *food, sustenance,* cp. OE. **fēdan,** Goth. **fōdjan,** *to feed*; Goth. **guþ-blōstreis,** *worshipper of God,* OHG. **bluoster,** *sacrifice,* cp. Goth. **blōtan,** *to worship.*

Instead of **ss** (**s**) we often meet with **st.** In such cases the **st** is due to the analogy of forms where **t** was quite regular, e. g. regular forms were Goth. **last,** *thou didst gather,* inf. **lisan**; Goth. **slōht,** *thou didst strike,* inf. **slahan**; OE. **meaht,** OHG. **maht,** *thou canst,* inf. **magan**; then after the analogy of such forms were made OE. **wāst** for ***wās,** Goth. **wáist** for ***wáis,** OHG. **weist** for ***weis,** *thou knowest*; OE. **mōst** for ***mōs,** *thou art allowed*; regular

forms were pret. sing. OE. **worhte,** Goth. **waúrhta,** OHG. **worhta,** beside inf. OE. **wyrcan,** Goth. **waúrkjan,** OHG. **wurken,** *to work*; then after the analogy of such forms were made OE. **wiste** beside **wisse,** OHG. **wista** beside **wissa,** *I knew*; OE. **mōste** for ***mōse** (= OHG. **muosa**), *I was allowed.*

For purely practical purposes the above laws may be thus formulated:—every labial + **t** became **ft**; every guttural + **t** became **ht**; and every dental + **t** became **ss, s** (**st**).

§ **241.** Prim. Germanic **ǥw,** which arose from Indg. **ǵh** (§ **237**) and from Indg. **q** (§ **238**) by Verner's law, became **ǥ** before **u,** in other cases it became **w,** as Goth. **magus,** *boy,* beside **mawi** from ***ma(ǥ)wī́,** *girl*; pret. pl. Anglian **sēgon** from ***sǣǥ(w)un,** *they saw,* beside pp. **sewen** from ***se(ǥ)wenós**; OE. **sīen** (**sīon, sēon,** cp. § **138**), Goth. **siuns,** from ***se(ǥ)wnís,** *a seeing, face*; OE. **snāw** (with **-w** from the oblique cases), Goth. **snáiws,** from ***snai(ǥ)waz,** prim. form ***snóiǵhos,** *snow*; OE. OHG. **snīwan** for ***snīgan,** formed from the third pers. sing. OE. **snīweþ,** OHG. **snīwit,** *it snows.* See § **239,** Note 1, § **249.**

§ **242.** Assimilation:— **-nw-** > **-nn-,** as OE. Goth. OHG. **rinnan** from ***rinwan,** *to run*; OE. **cinn,** Goth. **kinnus,** OHG. **kinni,** from ***genw-,** Gr. γένυ-ς, *chin, cheek*; Goth. **minniza,** OS. **minnira,** OHG. **minniro,** from ***minwizō,** *less,* cp. Lat. **minuō,** Gr. μινύθω, *I lessen*; OE. **þynne,** O.Icel. **þunnr,** OHG. **dunni,** *thin,* cp. Skr. fem. **tanví,** *thin.*

**-md-** > **-nd-,** as OE. Goth. OS. **hund,** OHG. **hunt,** prim. form ***kmtóm,** *hundred*; OE. **scamian,** Goth. **skaman,** OHG. **scamēn,** *to be ashamed,* beside OE. **scand,** Goth. **skanda,** OHG. **scanta,** *shame, disgrace.*

**-ln-** > **-ll-,** as OE. **full,** Goth. **fulls,** Lithuanian **pìlnas,** prim. form ***plnós,** *full*; OE. **wulle,** Goth. **wulla,** OHG. **wolla,** Lithuanian **wìlna,** *wool.*

§ **243.** Prim. Germanic **b̄n, đn, ǥn** = Indg. **pn̥, tn̥, kn̥, qn̥** (by Verner's law), and **bhn̥, dhn̥, ghn̥, ǵhn̥,** became

**ƀƀ, đđ, gg** before the principal accent, then later **bb, dd, gg**; and in like manner Indg. **bn̥, dn̥, gn̥, ǥn̥** became **bb, dd, gg.** And these mediae were shifted to **pp, tt, kk** at the same time as the original Indg. mediae became tenues (§ **232**). These geminated consonants were simplified to **p, t, k** after long syllables. Examples are: OE. **hnæpp,** OHG. **napf,** from ***χnaƀn̥** or **χnabn̥**, *basin, bowl*; OE. **hoppian,** O.Icel. **hoppa,** MHG. **hopfen,** from ***χoƀn̥**, *to hop*; OE. OS. **topp,** O.Icel. **toppr,** from ***toƀn̥** or **tobn̥**, *top, summit*; OE. **hēap,** OS. **hōp,** OHG. **houf,** from ***χauƀn̥**; OE. **cnotta,** from ***knođn̥**, beside OHG. **chnodo, chnoto,** *knot*; OE. OS. **hwīt,** Goth. **hveits,** from ***χwīđn̥**, *white*; OE. **bucc,** O.Icel. **bokkr,** OHG. **boc** (gen. **bockes**), prim. form ***bhugnós,** *buck*; OE. **liccian,** OS. **leccōn,** OHG. **lecchōn,** from ***legn̥**, *to lick*; OE. **locc,** O.Icel. **lokkr,** prim. form ***lugnós,** *lock*; OE. **smocc,** O.Icel. **smokkr,** from **smogn̥**, *smock*; OE. **lōcian,** OS. **lōkōn,** from **lōgn̥** or **lōgn̥**, *to look.*

§ **244.** Indg. **z** + media became **s** + tenuis, as Goth. **asts,** OHG. **ast** = Gr. ὄζος, from ***ozdos,** *branch, twig*; OE. OHG. **nest,** Lat. **nīdus,** from ***ni·zdos,** *nest,* related to root **sed·,** *sit*; OE. **masc,** OHG. **masca,** *mesh, net,* cp. Lithuanian **mezgù,** *I tie in knots.*

Indg. **z** + media aspirata became **z** + voiced spirant, as OE. **meord,** Goth. **mizdō,** *pay, reward,* cp. O. Bulgarian **mĭzda,** Gr. μισθός, *pay*; OE. **mearg,** OHG. **marg,** O. Bulgarian **mozgŭ,** *marrow,* root **mezgh·**; OE. **hord,** Goth. **huzd,** OHG. **hort,** *hoard, treasure,* root **kuzdh·.**

§ **245.** Guttural **n** (ŋ) disappeared before **χ**, as Goth. OS. OHG. **fāhan,** OE. **fōn,** from ***faŋχanan,** *to seize*; Goth. OS. OHG. **hāhan,** OE. **hōn,** from ***χaŋχanan,** *to hang*; Goth. **þeihan,** OS. **thīhan,** OHG. **dīhan,** OE. **þīon, þēon,** from ***þiŋχanan,** *to thrive*; pret. OE. **þōhte,** Goth. **þāhta,** OS. **thāhta,** OHG. **dāhta,** from ***þaŋχtō·,** *I thought,* beside inf. OE. **þencan.** See §§ **40–1.**